

Low-Temperature Germanium Waveguides for Mid-Infrared Sensing Applications

by

Erin Wei Zhang

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Authored by: Erin Wei Zhang

Department of Electrical Engineering and Computer
Science

August 15, 2025

Certified by: Lionel C. Kimerling

Thomas Lord Professor of Materials Science and
Engineering

Thesis Supervisor

Accepted by: Katrina LaCurts

Chair

Master of Engineering Thesis Committee

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Abstract

Waveguide integrated devices that operate in the mid-infrared (mid-IR) wavelength range (2.5-12 μm) are used for sensing the fundamental absorption bands in a variety of molecules. Germanium (Ge) is commonly used for photodetection in the near-infrared (near-IR) wavelength range of 1.2-1.6 μm due to its strong absorption from a 0.8 eV direct band gap. At longer wavelengths in the mid-IR range, Ge exhibits transparency that makes it a desirable waveguide material for sensing applications. Its epitaxial growth compatibility with silicon (Si) substrates makes Ge-on-Si an effective platform for mid-IR waveguides. For back-end-of-line (BEOL) integration of waveguides in sensing applications, the thermal budget limits the temperature to below 450°C. In this work, we investigated the use of h-line exposure as a commercially viable, low-cost option for patterning low temperature (LT) Ge-on-Si waveguides using direct write lithography. Waveguide dimensions for optimal confinement in single-mode transverse electric (TE) polarization at wavelengths of 3 μm and 10.4-11.3 μm were modeled and the direct lithography process was refined. Through dose testing and adjustments to the raster direction and pixel resolution, it was found that direct write lithography lacked the resolution required for low-loss waveguides. Scanning electron microscopy (SEM) revealed inconsistent waveguide widths and side-wall roughness, and e-beam lithography was identified as the preferred lithography process. For future integration of LT-Ge in a foundry process design kit (PDK), a universal thickness of 1.7 μm was found to support single-mode waveguide operation from 3-11.3 μm wavelength.

Thesis Supervisor: Lionel C. Kimerling

Title: Thomas Lord Professor of Materials Science and Engineering

Contents

1	Introduction	13
1.1	Mid-IR Benefits of Ge-on-Si Waveguides	14
1.2	Existing Ge-on-Si Waveguides	16
1.3	Ge-on-Si Epitaxy	18
1.4	Low-Temperature Germanium	18
2	Modeling Waveguide Modes	21
2.1	3 μm Wavelength Modeling	22
2.2	10.4–11.3 μm Wavelength Modeling	25
3	Process Refinement for LTGe-on-Si Waveguides	33
3.1	Waveguide Fabrication Process for 3 μm Wavelength	33
3.2	Waveguide Fabrication Process for 10.4–11.3 μm Wavelength	42
3.3	3 μm Wavelength Loss Measurement Setup	43
3.4	Summary and Next Steps	46

List of Figures

1-1	Materials transparency window with an optical loss below 2 dB/cm shows that germanium is transparent further into the mid-IR wavelength range than other common waveguide materials [1]	14
1-2	Methane absorption peak at 3.31 μm capable of ppm concentration sensitivity [2]	15
1-3	Diseases and their associated biomarkers in exhaled breath (EB) and exhaled breath condensate (EBC)	15
2-1	Single TE vs single TM mode	22
2-2	SEM image of 0.42 μm thick Ge-on-Si wafer from MIT Lincoln Labs .	23
2-3	Linear and log scale views of single TE mode at 2.08 μm x 0.42 μm .	24
2-4	Linear and log scale views of single TE mode at 1.22 μm x 0.50 μm .	24
2-5	Linear and log scale views of single TE mode at 1.08 μm x 0.54 μm .	24
2-6	10.4 μm wavelength single TE mode at a 2x1 aspect ratio	26
2-7	10.4 μm wavelength single TE mode at a 3x1 aspect ratio	27
2-8	10.9 μm wavelength single TE mode at a 2x1 aspect ratio	28
2-9	10.9 μm wavelength single TE mode at a 3x1 aspect ratio	29
2-10	11.3 μm wavelength single TE mode at a 2x1 aspect ratio	30
2-11	11.3 μm wavelength single TE mode at a 3x1 aspect ratio	31

3-1	GDS design for 3 μm wavelength using "paperclip" bends to vary waveguide lengths	34
3-2	The resulting waveguides post-development at a dosage of 100 mJ/cm	35
3-3	Cleanly etched waveguides	36
3-4	15 μm to 2.08 μm taper region of an etched waveguide with the micromasking effect; ion reflection from the sidewalls likely explains why the effect avoids the waveguide.	38
3-5	Cleanly cleaved 15 μm facet under SEM showing the overetch into the Si substrate and slanted sidewall	39
3-6	Poorly cleaved 15 μm facet under SEM	40
3-7	2 μm -wide waveguide SEM image showing how leftover photoresist or nonlinear scribing could be impacting the facet after cleaving	40
3-8	Sidewall roughness on the 2 μm -wide waveguide under SEM	41
3-9	An above view of the shadow from the sidewall angle	41
3-10	E-beam lithography after the development step shows smoother sidewall edges and less shadow from a sidewall angle compared to Figure 3-9.	42
3-11	Wafer 1 with surface defects	43
3-12	4 separate etch steps were carried out to try to minimize any wafer overheating. These images show the dose testing after the first and last etch, respectively, on 2 μm -thick Ge. The micromasking effect's damage is especially evident on the final result where almost the entire wafer has turned into "black-Si/Ge."	44
3-13	Waveguide next to Ge/Si pillars under microscope	45
3-14	Significantly improved wafer lot for 2 μm thick Ge on Si substrate	45
3-15	The loss measurement setup for the 3 μm wavelength laser	46

3-16	The single TM mode of a modeled strip waveguide with a width of 0.6 μm and a height of 1.7 μm	48
3-17	The single TE mode of a modeled strip waveguide with a width of 6 μm and the same height of 1.7 μm as Figure 3-15	49

List of Tables

1.1	Comparison of Mid-IR Ge-on-Si waveguides. The wavelength, loss, Ge thickness, and other platform parameters are included to provide a reference of existing related work. It is important to note that all the waveguides below are either explicitly high-temperature epitaxy or implied high-temperature due to a total dislocation density (TDD) of approximately 10^7 cm^{-2}	17
2.1	Dimension simulations at 3 μm wavelength	23
2.2	Bend simulations at 3 μm wavelength	25
2.3	Waveguide simulations at 10.4–11.3 μm wavelength	25
2.4	MIT Lincoln Labs wafer lot	32
3.1	The process flow after dose testing and other refinement	35

Chapter 1

Introduction

The mid-infrared (IR) range, spanning wavelengths from 2.5–12 μm , represents one of the most important domains for molecular sensing and identification. This wavelength range is often called the molecular fingerprint region because of the presence of fundamental vibrational absorption bands that are highly specific to molecular structure, enabling the precise identification and quantification of a wide variety of substances.

The bulk of research in photonic integrated circuits (PICs) has historically focused on silicon-on-insulator (SOI) platforms in the near-IR telecommunication band (1.3–1.55 μm wavelength). However, for sensing applications, the near-IR's overtone and combination bands exhibit significantly weaker absorption coefficients than the mid-IR's fundamental bands, making it more difficult to detect trace concentrations [3]. Additionally, in the mid-IR range, the existing SOI platforms become impractical due to strong absorption losses in silicon dioxide (SiO_2) above 3.5–4.0 μm wavelength [4]. Ge-on-SOI platforms are compatible with standard CMOS processing, facilitating large-scale integration of photonic devices such as photodetectors, modulators, and multiplexers on a single chip [5]. They also provide better thermal stability and electrical isolation due to the underlying SiO_2 buffer. However, the buried oxide

layer is absorbing beyond 3.7 μm wavelength, making Ge-on-SOI waveguides often impractical for the mid-IR range [6].

The limitations of existing platforms have driven interest in germanium-on-silicon (Ge-on-Si), which offers several critical advantages for mid-IR photonics. A BEOL compatible process in particular would significantly extend the existing capabilities of Ge-on-Si waveguides.

1.1 Mid-IR Benefits of Ge-on-Si Waveguides

Germanium provides an exceptionally wide transparency window, seen in Figure 1-1, unlike existing Ge-on-SOI platforms where the buried oxide layer is absorbing beyond 3.7 μm wavelength [1]. Ge-on-Si's transparency window spans key molecular "fingerprint" regions (3–12 μm) critical for sensing applications [7]. This transparency, combined with germanium's higher refractive index (approximately 4.0 at 4 μm wavelength) compared to silicon (approximately 3.4), enables strong optical confinement in compact waveguide geometries.

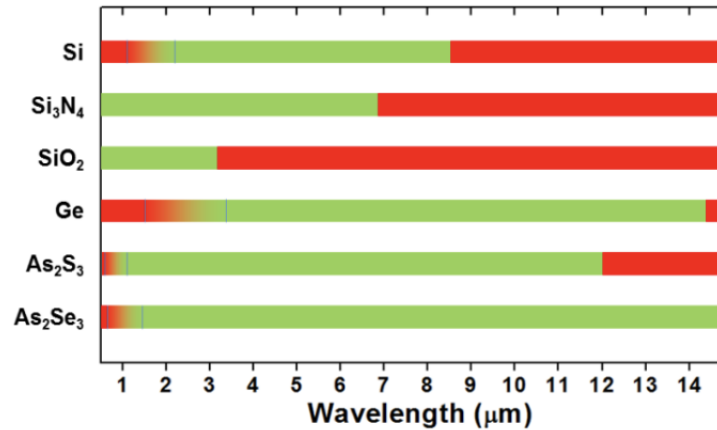


Figure 1-1: Materials transparency window with an optical loss below 2 dB/cm shows that germanium is transparent further into the mid-IR wavelength range than other common waveguide materials [1]

The high index contrast facilitates tight waveguide bends with minimal losses

and enables dense photonic integration that is critical for practical sensing devices. Additionally, Ge's compatibility with epitaxial growth on Si substrates allows for the fabrication of high-quality crystalline layers with reduced defects and controlled material properties, including precise doping profiles when required for active device applications [8]. Figure 1-2 shows an example of how Ge-on-Si PICs can be useful for environmental monitoring in the mid-IR, where greenhouse gases like methane show absorption peaks. Applications also exist for biomedical diagnostics, such as mid-IR sensors for exhaled breath to find "biomarkers for specific diseases, physiologic status, or therapeutic progress," seen in Figure 1-3 [9].

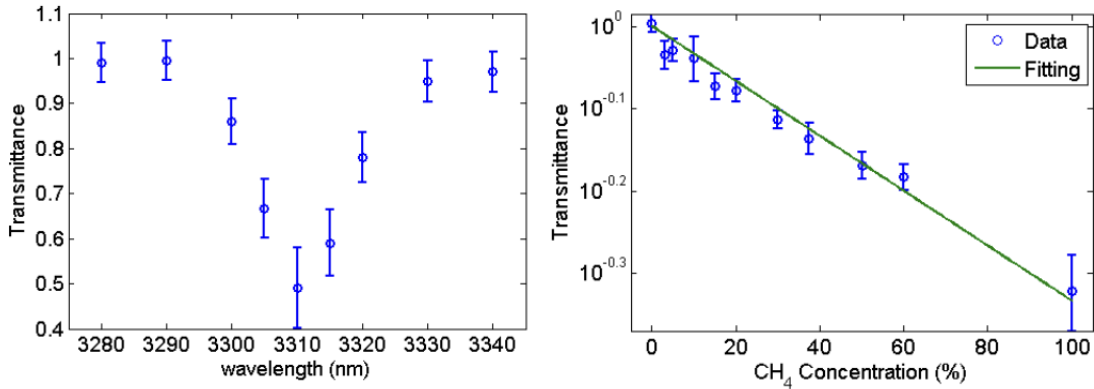


Figure 1-2: Methane absorption peak at 3.31 μm capable of ppm concentration sensitivity [2]

Disease / Health status	Biomarkers in EB	Biomarkers in EBC	Reference
Breast Cancer	BMAC, pentane, formaldehyde		[37, 38]
Lung Cancer	BMAC, NO, acetaldehyde	H ₂ O ₂ , nitrite	[39, 40, 133]
Asthma	NO, CO	H ₂ O ₂ , leukotriene E ₄ , isoprostanes, TBARs*	[27, 32, 134-138]
COPD	NO, CO	H ₂ O ₂ , leukotrienes, prostanoids, isoprostanes	[28, 33, 139, 140] [42, 141]
CF	NO ₂ , CO	H ₂ O ₂ , isoprostanes	[28, 34, 142]
ARDS	NO	H ₂ O ₂	[143-145]
Diabetes	Acetone		[8, 12, 14, 16]
Schizophrenia	CS ₂ , pentane		[35, 36]
Renal Function	NH ₃ , dimethylamine, trimethylamine		[146, 147]
Liver function, Liver transplant rejection	COS**		[148, 149]
Helicobacter Pylori infection	¹³ C-urea breath test		[150-154]

*thiobarbituric acid-reactive products (TBARs); ** carbonyl sulfide (COS)

Figure 1-3: Diseases and their associated biomarkers in exhaled breath (EB) and exhaled breath condensate (EBC)

1.2 Existing Ge-on-Si Waveguides

Recent research has demonstrated the significant potential of Ge-on-Si waveguides for mid-IR applications, with performance metrics that vary considerably depending on growth conditions, waveguide geometry, and operational wavelength. Notably, all of the mid-IR Ge waveguides below involve a high temperature step in the Ge epitaxy process.

The most impressive mid-IR results have been achieved by the University of Southampton, demonstrating propagation losses as low as 0.58 ± 0.12 dB/cm at $3.8 \mu\text{m}$ using $2.9 \mu\text{m}$ thick germanium rib waveguides [10]. These devices were fabricated using reduced-pressure chemical vapor deposition (RPCVD) with threading dislocation densities (TDD) of $2\text{--}5 \times 10^7 \text{ cm}^{-2}$. Similar performance was reported by MIT Lincoln Laboratories, achieving losses of 0.61 ± 0.1 dB/cm for TE polarization and 0.59 ± 0.1 dB/cm for TM polarization at $4.6 \mu\text{m}$ wavelength using $1 \mu\text{m}$ thick germanium [11]. Other research groups have reported higher losses but still achieved practically useful performance. Nanyang Technological University demonstrated 8.18 ± 0.6 dB/cm at $3.8 \mu\text{m}$ wavelength using a standard high-temperature three-step growth method [12], while research at Ghent University achieved 2.5-3.5 dB/cm in the $5.15\text{--}5.4 \mu\text{m}$ wavelength range using atmospheric pressure chemical vapor deposition (APCVD) with high-temperature annealing [13].

At longer wavelengths, the University of Glasgow has reported some of the most promising results, achieving losses below 1 dB/cm at $10\text{--}11 \mu\text{m}$ wavelengths using $2 \mu\text{m}$ thick germanium rib waveguide [14]. These devices employed thermal cyclic annealing to reduce the TDD to $\leq 10^7 \text{ cm}^{-2}$.

Table 1.1: Comparison of Mid-IR Ge-on-Si waveguides. The wavelength, loss, Ge thickness, and other platform parameters are included to provide a reference of existing related work. It is important to note that all the waveguides below are either explicitly high-temperature epitaxy or implied high-temperature due to a total dislocation density (TDD) of approximately 10^7 cm^{-2} .

Reference	Wavelength (μm)	Loss (dB/cm)	Ge thickness (μm)	Polarization	Platform	Growth Method	Highest Temperature ($^{\circ}\text{C}$)
Nedeljkovic @ Southampton	3.8	0.59 ± 0.12	2.9	TE	Rib waveguide Width = 2.7 μm and etch depth = 1.7 μm E-beam lithography, dry etched using SF_6 and C_4F_8 gases	RPCVD; TDD = $2-5 \times 10^7 \text{ cm}^{-2}$	850
Li @ Nanyang Technological University	3.8	8.18 ± 0.6	1	TE	Strip waveguide 800 μm taper from 20 μm to 2 μm	RPCVD Single crystal Ge with three-step Ge growth method	850
Malik @ Ghent University	5.15-5.4	2.5-3.5 3-4	2	TE TM	Ti/Cr metal mask - dry etching of germanium is carried out in a CF_4/O_2 plasma, followed by an HF dip to remove the metal mask AWG with 200 GHz (18 nm) channel spacing with five channels (146 nm free spectral range) I-line contact lithography	APCVD (atmospheric) TDD = 10^7 cm^{-2}	850
Gallacher University of Glasgow	5.8	~ 2	2		Rib waveguide <4.25 μm wide, 1 μm deep Grating coupler and rib waveguides were dry etched at the same time using a mixed gas process (SF_6 / C_4F_8)	Low-energy PECVD Cyclic annealed at high temperature to reduce the threading dislocation density to $\sim 10^7 \text{ cm}^{-2}$	"Cyclic annealed at high temperature"
Chang	5.8	2.5-3.0	2	TM	Strip waveguide; 500 μm -long coupling elements, 350 μm -long taper 2.9 μm wide Fluorine-based RIE Monocrystalline germanium layer on a silicon substrate with a 10 μm -thick intrinsic silicon buffer layer	RPCVD 2-step HT-Ge growth, relaxed Ge TDD = $5 \times 10^7 \text{ cm}^{-2}$	750-890
Fong @ A*STAR Singapore	5.5-10.7 >9	4.2-37 >15	3		1 μm Si overetch	RPCVD	
Mashanovich and Nedeljkovic @ Southampton	7.5	3.0 ± 0.8	3	TE	Rib waveguide Depth = 1.7 μm and width = 4.3 μm E-beam and ICP etching Grating couplers	RPCVD TDD = $2-5 \times 10^7 \text{ cm}^{-2}$	
Gallacher University of Glasgow	10-11 7.5-11	<1 <5	2	TE, TM	Rib waveguide 4 μm wide, 1 μm deep Etched using SF_6 and C_4F_8 Taper from 12.5 μm	RPCVD Two temperature technique with thermal cyclic annealing, TDD $\leq 10^7 \text{ cm}^{-2}$	500 followed by anneal step

1.3 Ge-on-Si Epitaxy

Presently, most germanium is grown at high temperatures. Standard technology for Ge-on-Si growth in research and industry uses Chemical Vapor Deposition (CVD) with gases like germane (GeH_4) as precursors to thermally activate growth. This approach produces misfit and threading dislocations due to lattice mismatch between Si and Ge and other strain [15]. In the near-IR wavelength range where Ge is an absorber and used as a photodiode, these dislocations have negative effects. To reduce the threading dislocations produced, a cyclic annealing step is often added with temperatures alternating between around 750°C and 900°C. But this process is not BEOL integration compatible, as research has determined 450°C to be the maximum low-temperature growth condition for BEOL integration [16].

In the mid-IR wavelength range where Ge is transparent and used as a waveguide, however, scattering losses from the dislocations decrease because Rayleigh scattering is proportional to λ^{-4} [17]. As a result, bypassing the high-temperature annealing step could still potentially produce low-loss waveguides.

1.4 Low-Temperature Germanium

A current low-temperature method for growing germanium on silicon is polycrystalline germanium (poly-Ge), particularly in the form of nanostrips embedded in polymer claddings, has recently attracted attention as a low-cost alternative for integrated photonic devices, especially in the mid-IR range. Instead of relying on complex epitaxial growth required for crystalline wafers, poly-Ge can be fabricated using straightforward deposition techniques such as e-beam evaporation [18].

Poly-Ge, when embedded in advanced polymer platforms, is a viable and scalable material for mid-IR photonic devices. Poly-Ge is generally less expensive to produce than single-crystal germanium, making it an attractive option for large-scale

or cost-sensitive applications. Poly-Ge can also be readily deposited onto silicon wafers using standard thin-film processes, and it is highly compatible with existing silicon semiconductor manufacturing infrastructure, making it useful for PICs [18]. However, Poly-Ge films inherently contain numerous grain boundaries and structural defects, leading to optical scattering and difficulties in fabrication [19]. Through the Fabry-Perot method, previously reported losses have been "between 12-19 dB/cm at 9.79 μm with 15 dB/cm average transmission loss for the 3.4-4.6 group index range" [20]. These resulting higher losses serve as a reference point of existing LT-grown Ge waveguides.

An alternative to poly-Ge is single-crystalline Ge, which has been grown at 400°C using Ultra-High Vacuum CVD (UHVCVD) to produce a low tensile strain of 0.05% and defect density of $0.86\text{-}1.72 \times 10^{11}\text{cm}^{-2}$ [21]. This defect density is comparable to pre-anneal high-temperature growth [22].

The modern state-of-the-art commercial process for epitaxy is performed at low or reduced pressures (LPCVD or RPCVD), and a BEOL-compatible process for Ge-on-Si was developed for photodiode applications. These low-temperature Ge-on-Si photodiodes in the near-IR range have been shown to perform similarly to high-temperature devices despite thermal budget limitations [16]. We have taken the initial steps toward analyzing the efficacy of this LT-Ge recipe in the mid-IR wavelength range. In doing so, the foundation for developing a new, commercially viable process using single-crystalline LT-Ge and a setup for evaluating waveguide designs have been established.

Chapter 2

Modeling Waveguide Modes

This thesis explores how Ge-on-Si waveguides can be leveraged for sensing applications in the mid-IR range. The work undertaken made use of Ansys Lumerical’s MODE Finite Difference Eigenmode (FDE) solver for waveguide modeling. Approximately 0.5 μm -thick and 2 μm -thick Ge layers on silicon substrate wafers from MIT Lincoln Laboratories will be used.

The goal of simulation was to determine the maximum waveguide dimensions for a strip waveguide to achieve an optimal balance between optical confinement and single-mode transverse electric (TE) operation. A strip waveguide design was initially selected for its straightforward modeling and fabrication. Single-mode TE was chosen to be the polarization method because TE modes exhibit a higher effective refractive index compared to TM modes, enabling stronger light confinement in the waveguide core, reducing radiative losses. TE mode polarized waveguides are thus able to support tighter bends at low losses when atop the Si substrate. For sensor applications where spiral structures are often used to improve sensitivity, tighter bends facilitate more efficient use of space. The field distribution of a TE mode compared to its TM counterpart, seen in Figure 2-1, also makes coupling applications more straightforward. Lastly, propagating only the TE_{00} mode helps to maintain the

polarization, reducing the dispersion and distortion that result in higher losses.

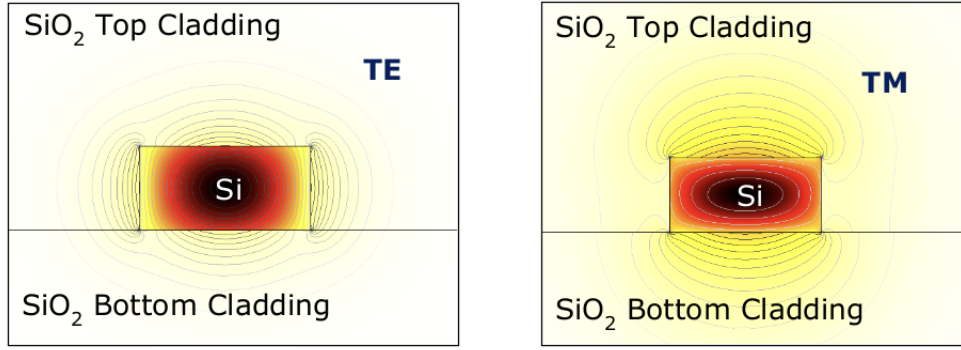


Figure 2-1: Single TE vs single TM mode

The anticipated testing method for these waveguide designs post-fabrication was to use 'paperclip' bends to vary waveguide lengths while keeping the number of bends constant, thus making loss measurements solely dependent on the waveguide length. As such, optimal bend radii to create minimal loss while preserving wafer space were also simulated for the 3 μm wavelength laser, the first to be tested. A suitable bend radius for this wavelength range and its associated waveguides was 250 μm .

2.1 3 μm Wavelength Modeling

We initially had only a 0.5 μm thick Ge-on-Si wafer from MIT Lincoln Labs available for processing and measurements. Slightly different thicknesses were also simulated in case of growth variation. Upon SEM imaging shown in Figure 2-2, the wafers from the first fabrication run were determined to be 0.42 μm thick, for which previous simulation established an optimal width of 2.08 μm . Figures 2-3 to 2-8 show linear and logarithmic scale images of the modeled modes at slightly varying thicknesses. It can be seen that even slight variations in the Ge layer thickness affects the maximum waveguide width and resulting mode significantly.

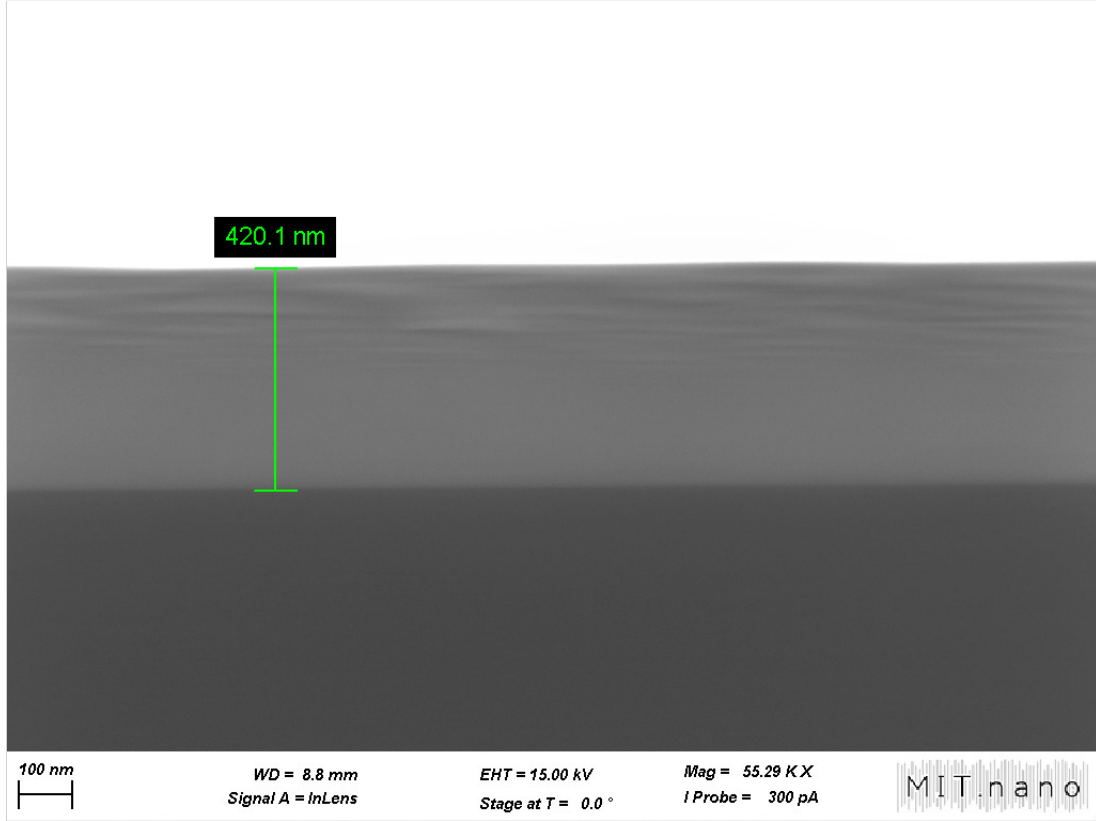


Figure 2-2: SEM image of 0.42 μm thick Ge-on-Si wafer from MIT Lincoln Labs

Table 2.1: Dimension simulations at 3 μm wavelength

Thickness (μm)	Width (μm)
0.42	2.08
0.44	1.78
0.46	1.52
0.48	1.38
0.50	1.22
0.52	1.10
0.54	1.08

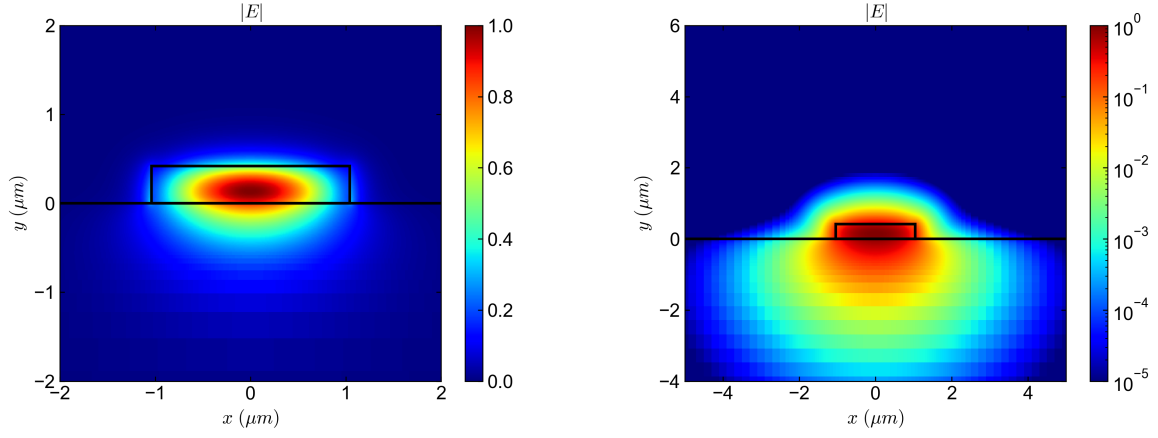


Figure 2-3: Linear and log scale views of single TE mode at $2.08 \mu\text{m} \times 0.42 \mu\text{m}$

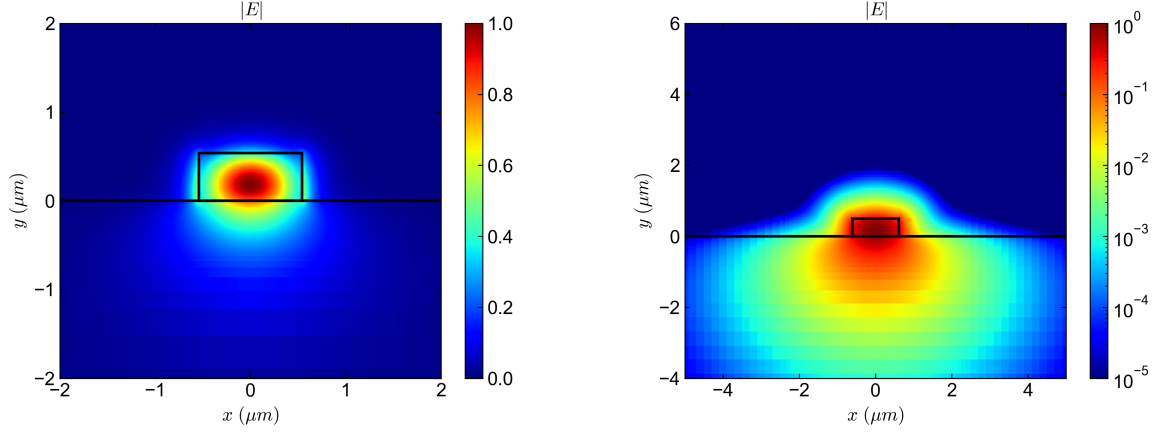


Figure 2-4: Linear and log scale views of single TE mode at $1.22 \mu\text{m} \times 0.50 \mu\text{m}$

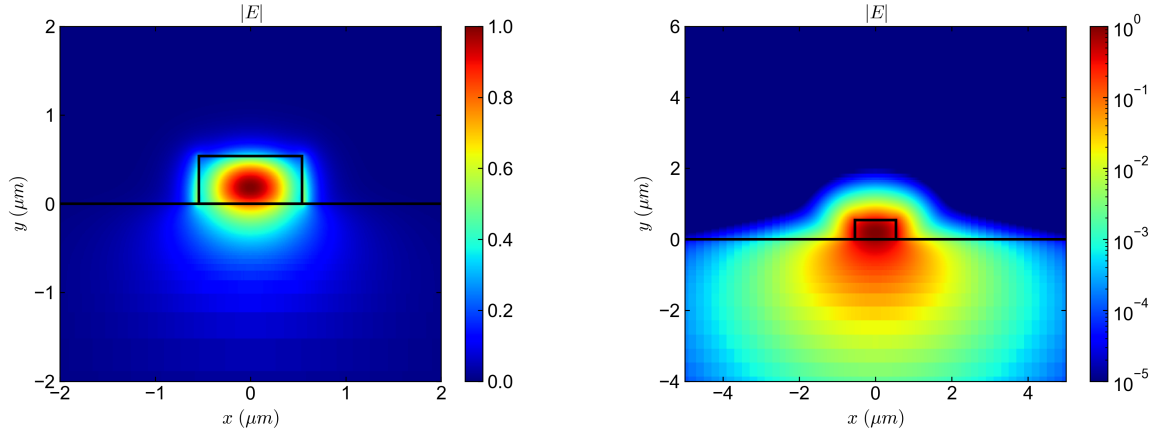


Figure 2-5: Linear and log scale views of single TE mode at $1.08 \mu\text{m} \times 0.54 \mu\text{m}$

Table 2.2: Bend simulations at 3 μm wavelength

Bend Radius (μm)	Overlap (%)	Loss (dB/cm)	Angular Loss (dB/rad)
150	99.63	1.81	0.027
170	99.60	1.52	0.026
190	99.87	0.185	0.0035
210	99.92	0.0505	0.0011
230	99.94	0.0276	0.00064

After the bend modeling seen in Table 2.2, we chose a bend radius of 250 μm to improve the bend efficiency even further beyond simulation without sacrificing significantly more wafer area. Although the measurements will be normalized across the bending losses, losses that are too high could potentially prevent too little light from coupling out and impact reliable data collection.

2.2 10.4–11.3 μm Wavelength Modeling

Unlike the modeling done for a 3 μm wavelength laser, the modeling for our longer wavelength laser was not based on an existing wafer thickness. Simulations were then done to determine an approximate range of thicknesses for which to request additional wafers from Lincoln Labs. We modeled the largest waveguide dimensions at two aspect ratios – 2x1 and 3x1 – that would only support a single TE mode across the wavelength range: at 10.4 μm , 10.9 μm , and 11.3 μm , seen in Table 2.3. The associated mode images are also shown in Figures 2-9 to 2-14.

Table 2.3: Waveguide simulations at 10.4–11.3 μm wavelength

Wavelength (μm)	2x1 Aspect Ratio Dimensions (width x height)	3x1 Aspect Ratio Dimensions (width x height)
10.4	3.94 x 1.97 μm	5.22 x 1.74 μm
10.9	4.14 x 2.07 μm	5.49 x 1.83 μm
11.3	4.28 x 2.14 μm	5.73 x 1.91 μm

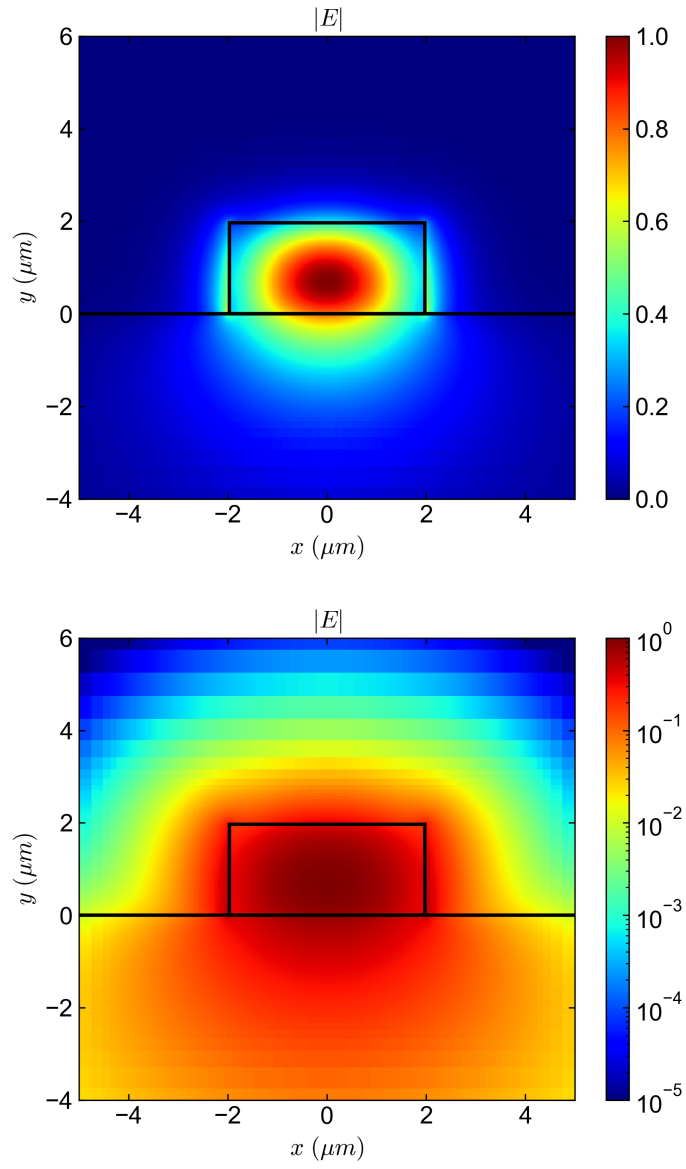


Figure 2-6: 10.4 μm wavelength single TE mode at a 2x1 aspect ratio

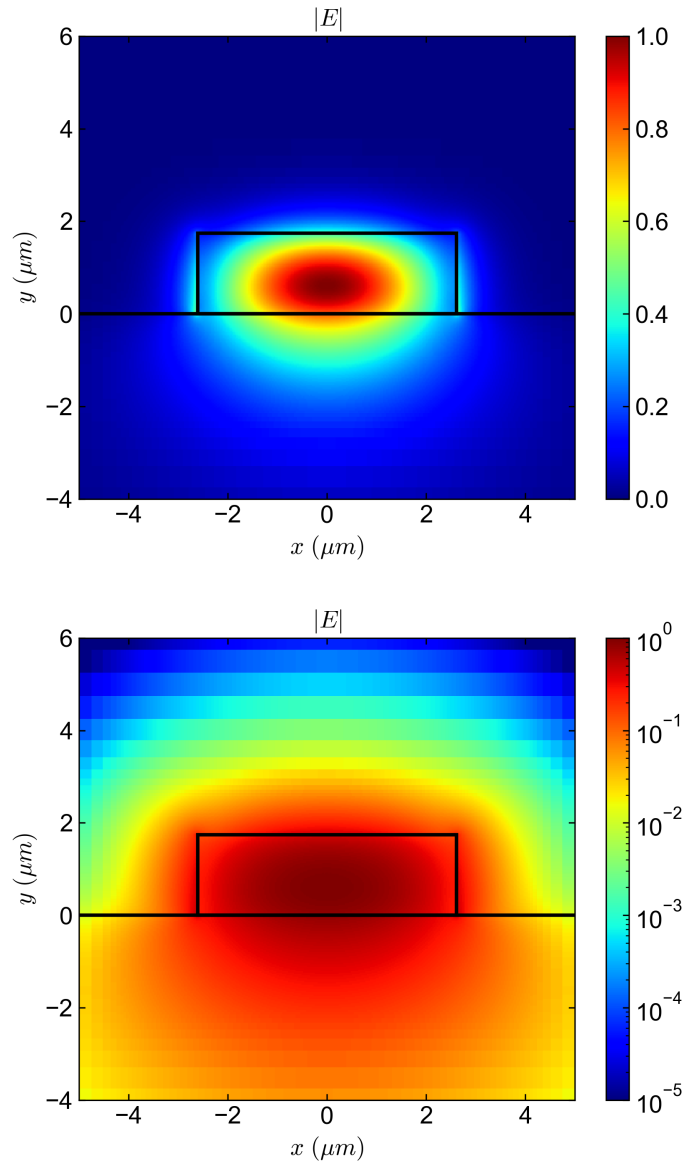


Figure 2-7: 10.4 μm wavelength single TE mode at a 3x1 aspect ratio

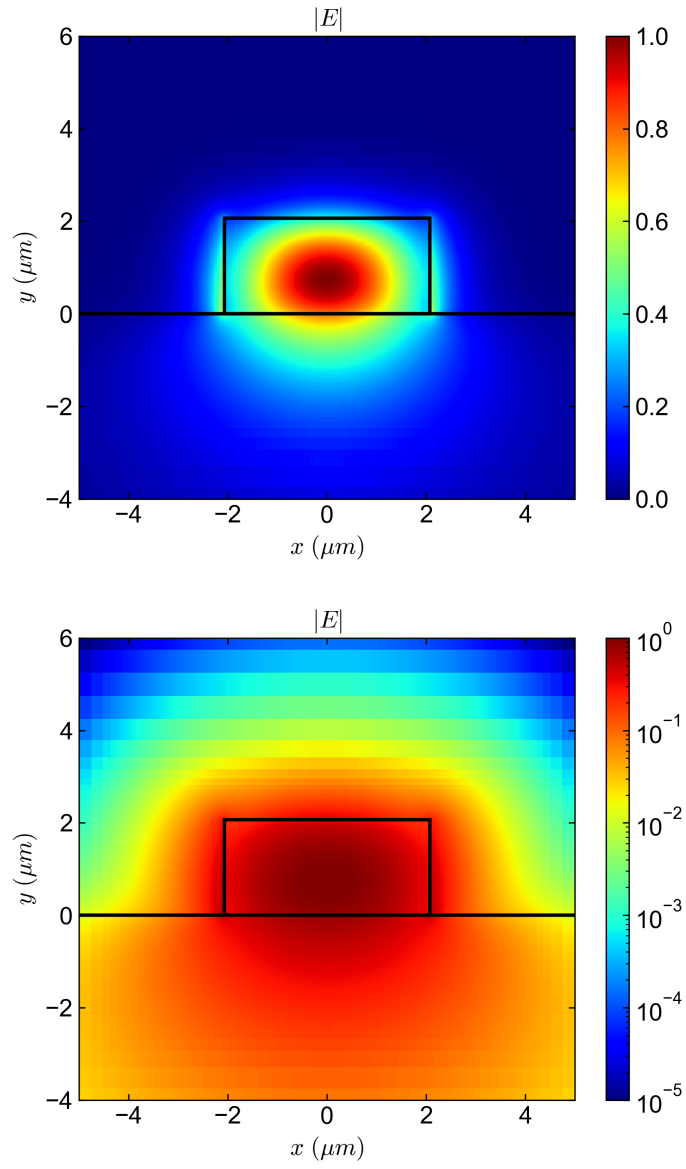


Figure 2-8: 10.9 μm wavelength single TE mode at a 2x1 aspect ratio

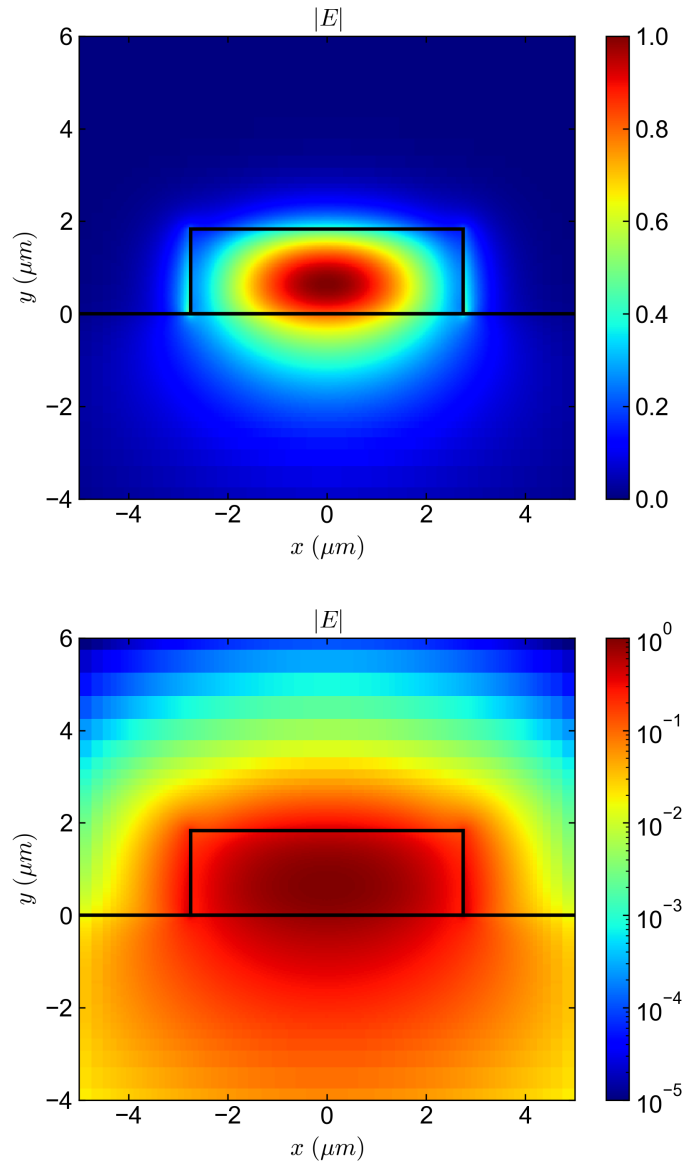


Figure 2-9: 10.9 μm wavelength single TE mode at a 3x1 aspect ratio

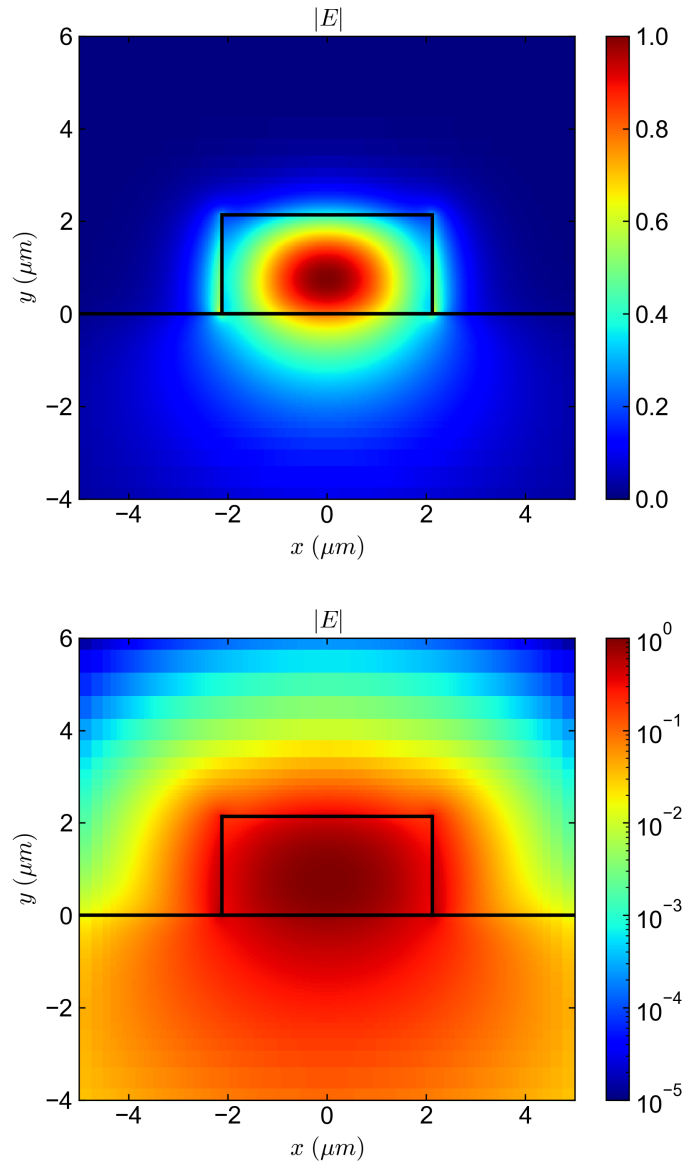


Figure 2-10: 11.3 μm wavelength single TE mode at a 2x1 aspect ratio

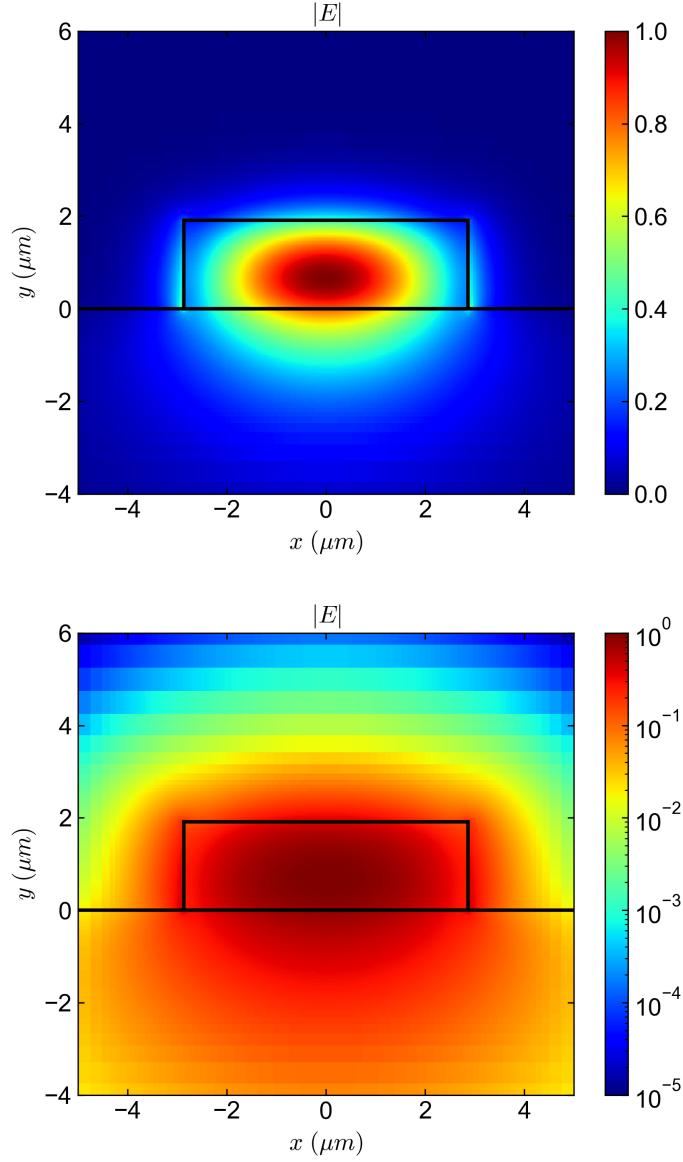


Figure 2-11: 11.3 μm wavelength single TE mode at a 3x1 aspect ratio

Through these simulations and the literature review seen in Table 1.1, we determined that the thicknesses of the Ge layer in the next Lincoln Labs LT-Ge growth run should be around 2 μm . The exact wafers requested on a standard Czochralski (CZ) silicon substrate can be seen in Table 2.4. Wafers 5 and 6 with no Si epitaxy can be compared to wafers 1 and 3, respectively, to evaluate the impact of the presence of oxygen in the Cz-Si substrate. Select wafers also included an SiO_2 atomic

layer deposition (ALD) for an initial test of losses due to SiO_2 absorption. This is to support an exploration into having strips of ALD on top of the Si substrate with Ge epitaxial overgrowth that coalesces into a uniform layer. If sufficiently low losses are found, the Ge-ALD-Si platform will be useful for further isolating the mode from the substrate and increasing the sensitivity for sensing applications.

Table 2.4: MIT Lincoln Labs wafer lot

Wafer #	Si Layer	SiO_2 ALD	ALD Thickness (nm)	Ge Thickness (μm)
1	1 μm	N	0	1.8
2	1 μm	N	0	1.9
3	1 μm	N	0	2.0
4	1 μm	N	0	2.1
5	N	N	0	1.8
6	N	N	0	2.0
7	1 μm	Y	1	1.8
8	1 μm	Y	5	1.8

Chapter 3

Process Refinement for LTGe-on-Si Waveguides

3.1 Waveguide Fabrication Process for 3 μm Wavelength

To begin the fabrication process for the 0.42 μm thick wafers we already acquired, a GDS waveguide design was created using the gdstk Python package. The waveguide design incorporated varying lengths from 1.3–3.1 cm in 0.2 cm increments. Each waveguide includes a standard taper from 15 μm to 2.08 μm , the width determined by simulation, in order to simplify the coupling process in testing. The number of bends and bend radius across each waveguide is also kept constant to 4 bends at 250 μm (also determined from simulation) to have normalized loss measurements.

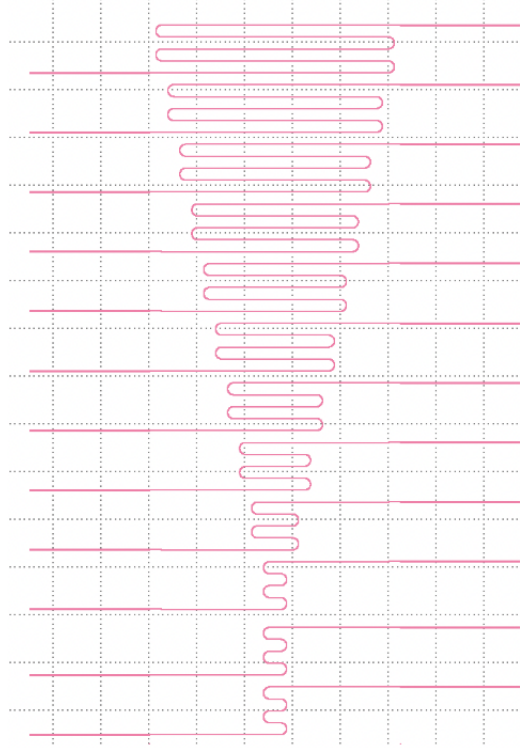


Figure 3-1: GDS design for 3 μm wavelength using "paperclip" bends to vary waveguide lengths

Following the waveguide design, the facilities at MIT.Nano were used to tailor a process for waveguides with clean definition and accurate width. This was done by testing both orientations for the Heidelberg DirectWrite-MLA150 lithography tool and dose testing. We found that the HexaMethylDiSilazane (HDMS) step prior to lithography was crucial to the resist adhesion. Without it, some waveguide resist patterns would lift completely and others would lift and no longer be straight once they adhered again. In the lithography step, having the laser raster direction going in the same direction as the waveguides yielded much more consistent waveguide widths and smoother sidewalls under a microscope. With such fine features, we also needed to experiment with developer technique and time in order to not overdevelop. The AZ726 MIF was used for its consistent develop time between the 15 μm - and 2 μm -wide features. At the end of all the process refinement, most of the resulting waveguides appeared cleanly fabricated under a microscope.

Table 3.1: The process flow after dose testing and other refinement

Step	MIT.Nano Tool	Description
0		Cleave wafer into pieces for processing
1	Branson Sonicator	Acetone, Methanol, IPA each for 3 min
2	HDMS-Yes-U10	Recipe 1
3	Spinner-DualResist-U10	AZ3312 @ 3000 RPM
4	Hotplate-Tower-U12	60s soft bake @ 100°C
5	DirectWrite-MLA150-AirAF	90-100 mJ/cm @ 405 nm
6	Hotplate-Tower-U12	60s PEB @ 110°C
7	Develop-U12	60s AZ726 MIF
8	RIE-CI-Samco-230iP	Dry etching for Ge on Si waveguides
9	Dektak-XT	Check etch depth
10	Asher-Barrel-Thierry/Asher-Chuck-ESI	Photoresist strip

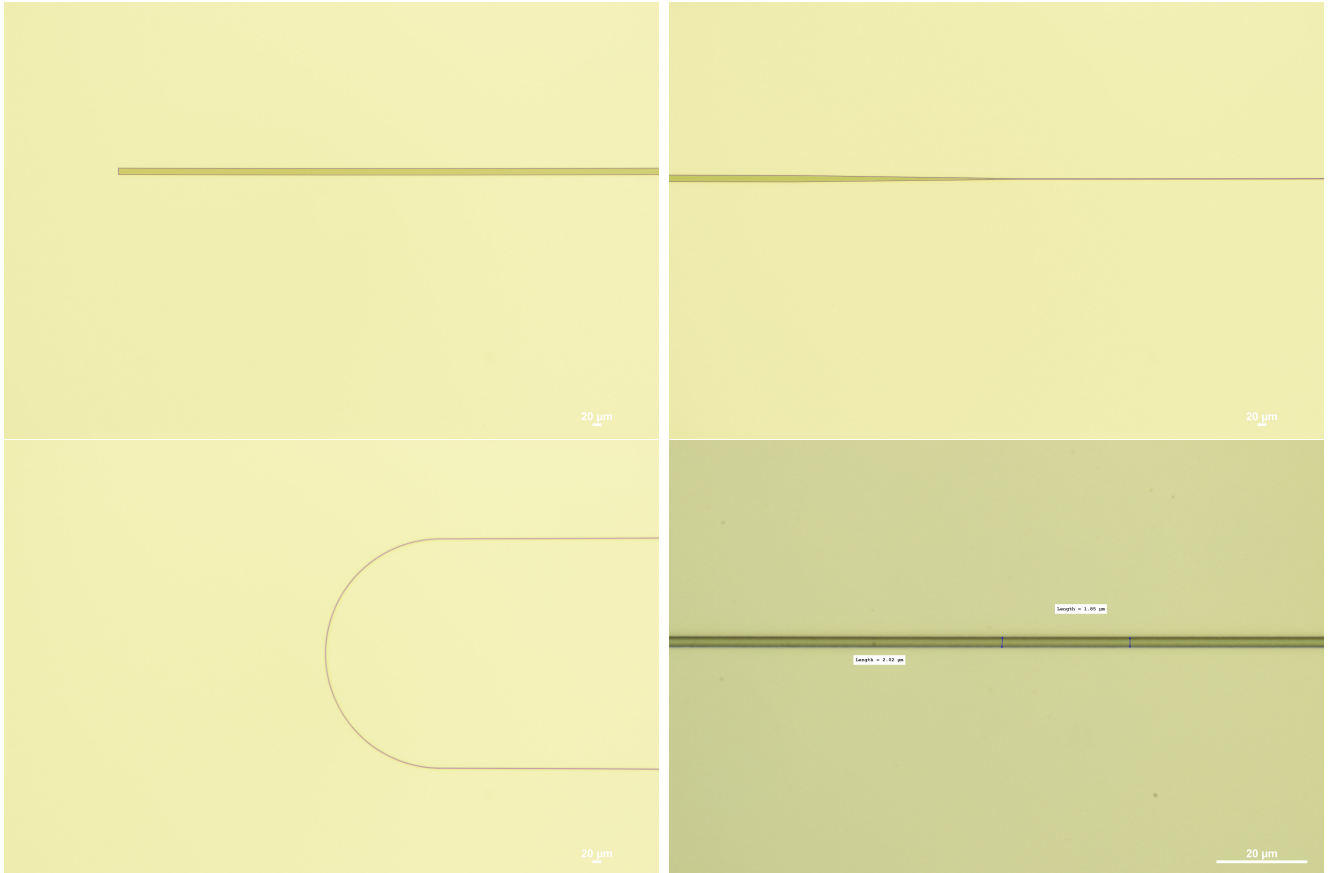


Figure 3-2: The resulting waveguides post-development at a dosage of 100 mJ/cm

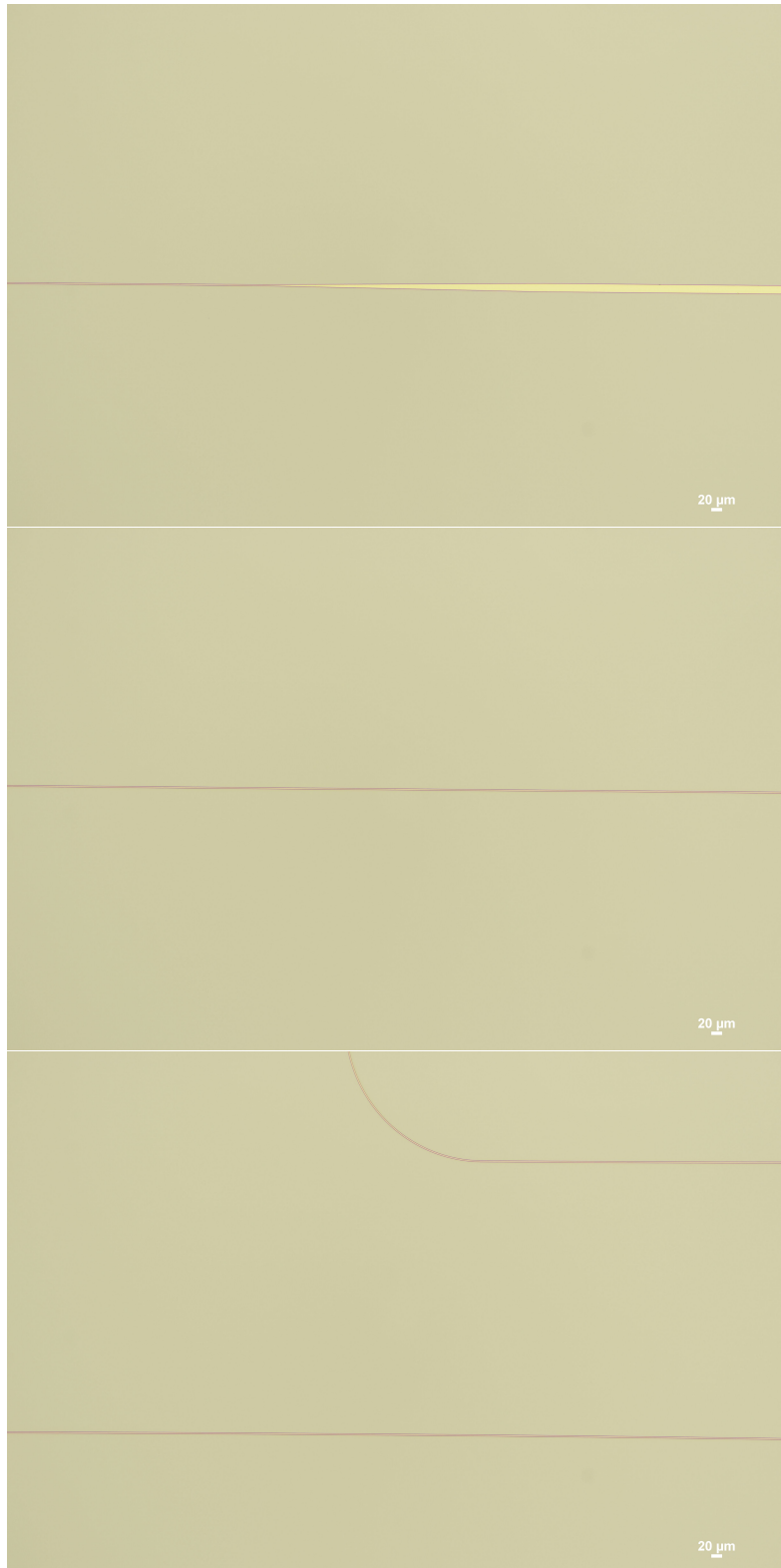


Figure 3-3: Cleanly etched waveguides

However, in the etching process using the tool's existing chlorine (Cl_2) recipe,

some parts of the wafer were left with what are likely pillars of Ge. This happens during the etch when residue forms or byproducts deposit and cause micromasking of the etch. Micromasking can also be caused by incomplete removal of the native oxide. The effect seems to disappear right next to the waveguide due to ion reflection from the sidewalls. The waveguide width was also less than the desired $2.08\text{ }\mu\text{m}$ at closer to $1.9\text{ }\mu\text{m}$ under the microscope. Nonetheless, simulation showed that a $1.9\text{ }\mu\text{m}$ x $0.42\text{ }\mu\text{m}$ waveguide would still support a single TE mode.

Despite what is likely micromasking, since most of the waveguides did not have these residue marks around them, we decided to proceed to cleave the wafer inside the $15\text{ }\mu\text{m}$ -wide region on both sides to expose the waveguides for coupling. The cleaves appeared to be cleanly along a plane under a microscope and we proceeded to test the completed waveguides, but later SEM imaging revealed additional issues shown in Figures 3-4 to 3-9. The scribing and cleaving produced inconsistent results, since crooked scribing could easily cause the wafer to not cleave along a facet, and any leftover photoresist could also make the cleaved waveguide face impossible to couple into. The etch process would not only cause micromasking but also over-etch and produce a slanted sidewall. Lastly, the lithography process yielded rough sidewalls that would cause high propagation losses.

The etch time could be reduced to prevent overetching and the ashing time could be increased (or Piranha solution - a mixture of concentrated sulfuric acid with hydrogen peroxide - used) to ensure complete photoresist removal. A hydrofluoric acid (HF) dip right before the etch would also remove the native oxide and other leftover residue to prevent micromasking. The cleaving, despite being inconsistent, could also be improved enough through alternative scribing methods to yield enough testable waveguides. However, the direct write lithography was already using the minimum spot size and preferred raster direction, indicating that the limit of the lithography method had been reached. Groups that had achieved successful low loss measure-

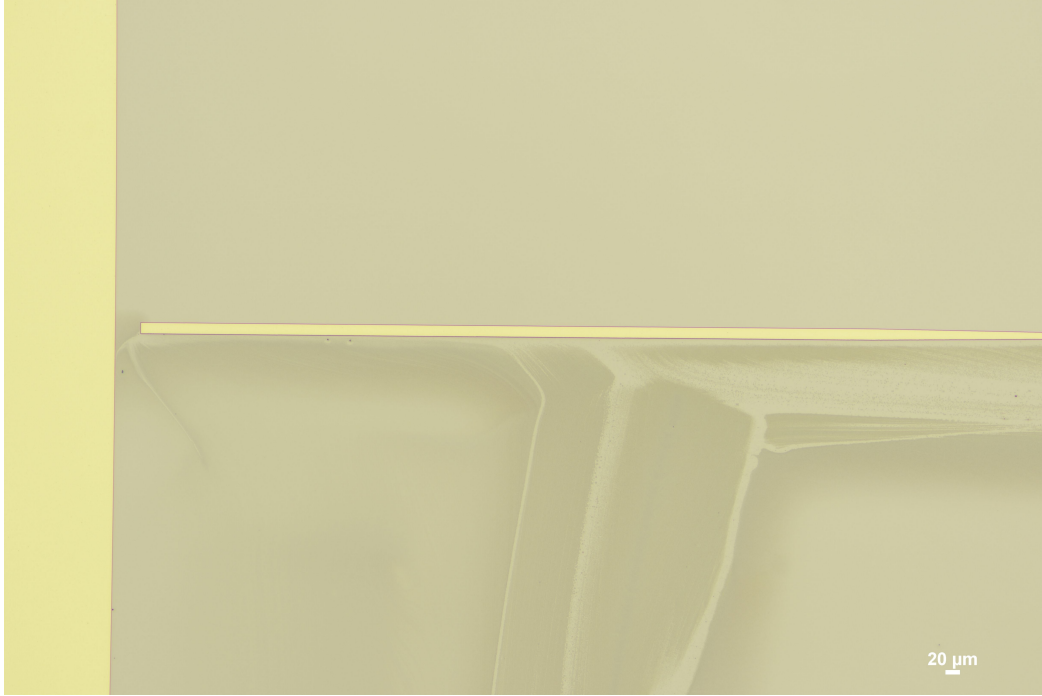


Figure 3-4: 15 μm to 2.08 μm taper region of an etched waveguide with the micro-masking effect; ion reflection from the sidewalls likely explains why the effect avoids the waveguide.

ments had used electron-beam (e-beam) lithography, so we decided to pivot to that as well.

An initial run prior to dose testing used AZ NLOF resist and did not include the HDMS prime step. Adhesion problems arose again upon development and, similarly to the direct write lithography, the waveguide widths were narrower than intended. However, the sidewalls appeared significantly straighter and smoother (see Figure 3-10). Further process refinement for fabricating these waveguides for a 3 μm laser should begin with e-beam lithography with an HSQ resist for further improved sidewall roughness and angle.

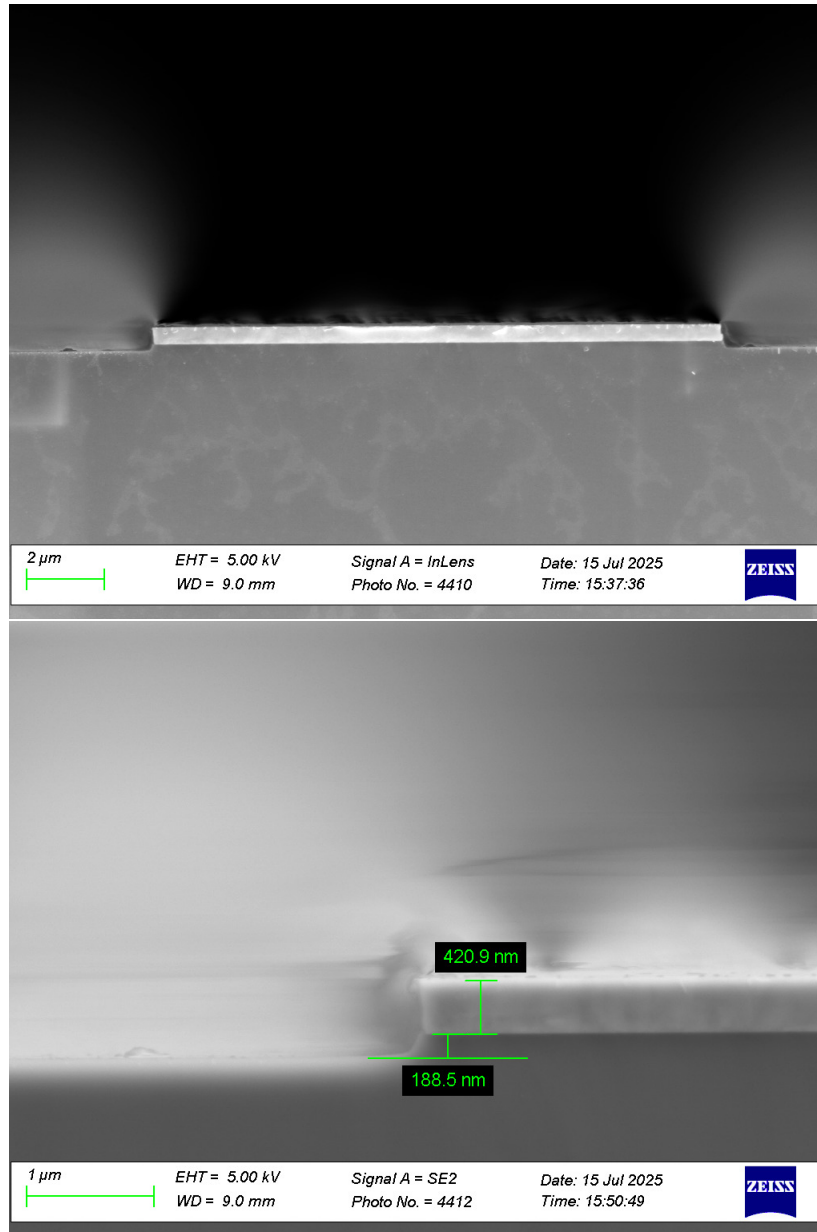


Figure 3-5: Cleanly cleaved 15 μm facet under SEM showing the overetch into the Si substrate and slanted sidewall

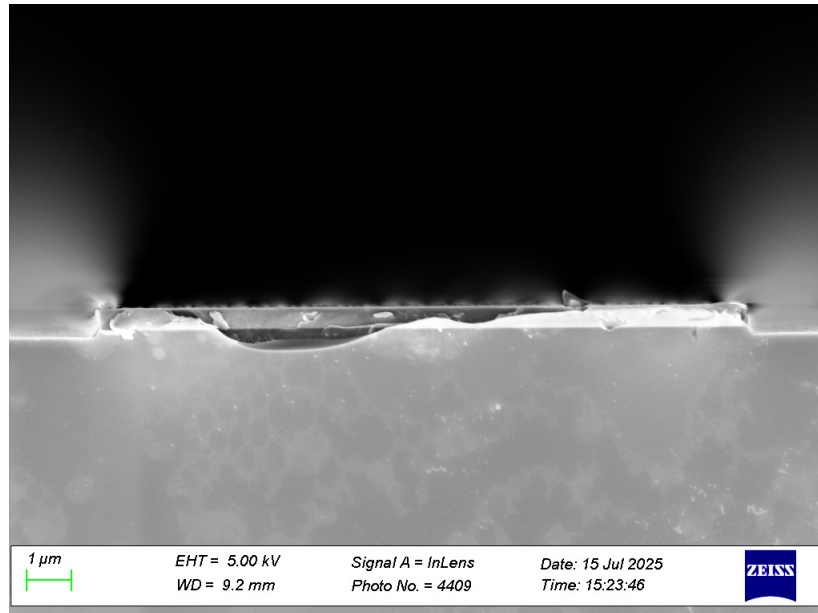


Figure 3-6: Poorly cleaved 15 μm facet under SEM

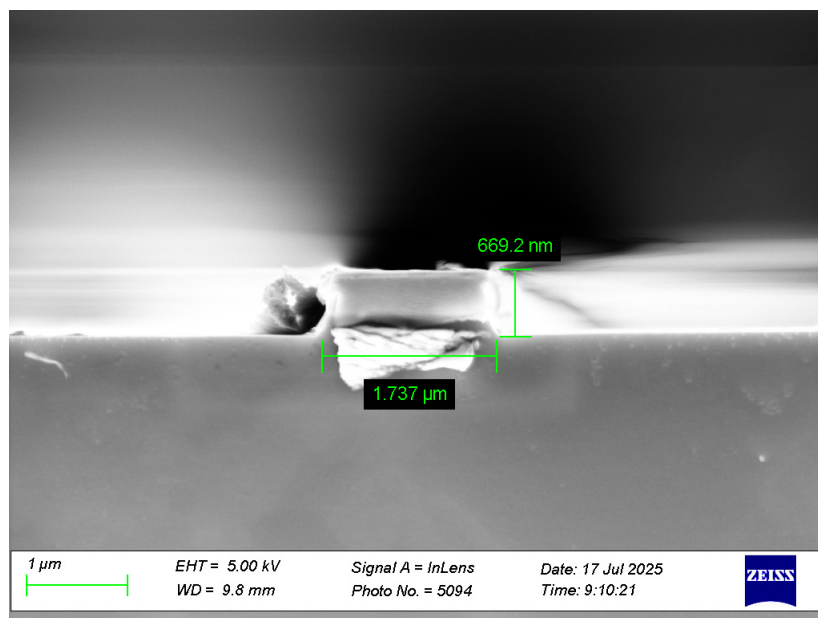


Figure 3-7: 2 μm -wide waveguide SEM image showing how leftover photoresist or nonlinear scribing could be impacting the facet after cleaving

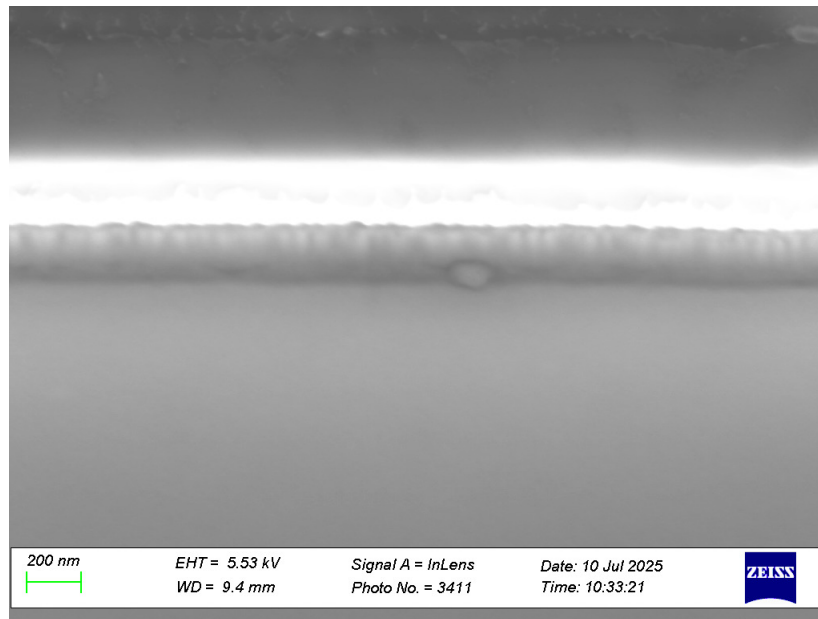


Figure 3-8: Sidewall roughness on the 2 μm -wide waveguide under SEM

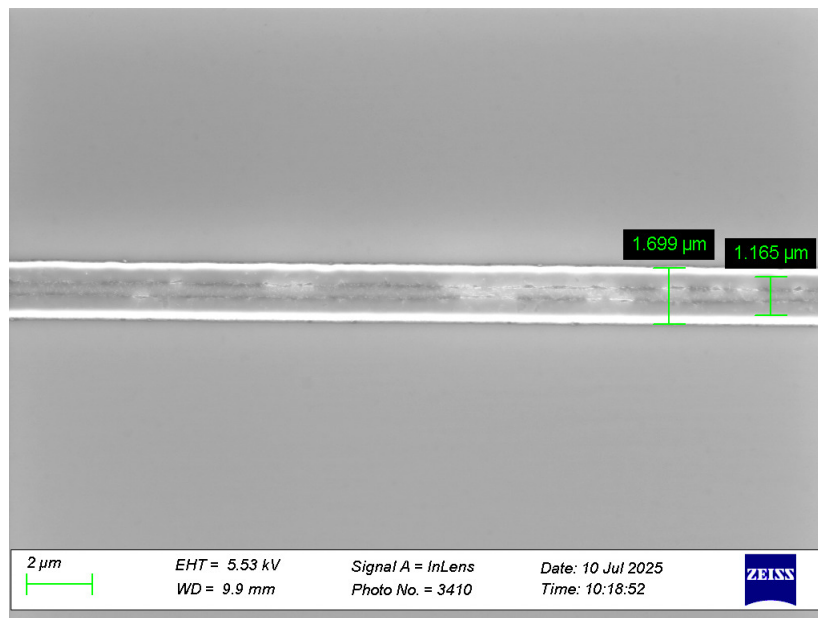


Figure 3-9: An above view of the shadow from the sidewall angle

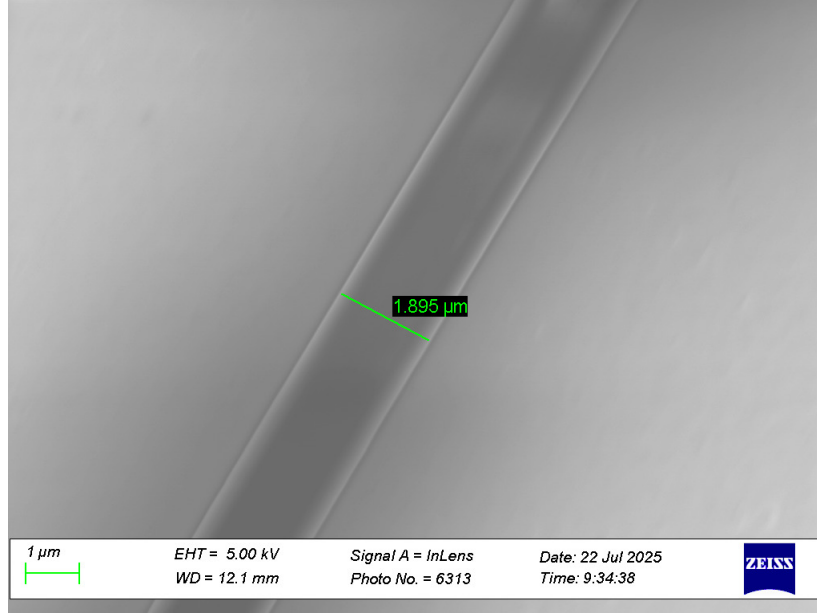


Figure 3-10: E-beam lithography after the development step shows smoother sidewall edges and less shadow from a sidewall angle compared to Figure 3-9.

3.2 Waveguide Fabrication Process for 10.4–11.3 μm Wavelength

Upon microscope imaging of the received wafers, we discovered that the growth had an abnormal amount of defects relative to previous LT-Ge wafers. The imperfections can be observed in Figure 3-11.

The wafers were still able to be used for dose testing on the direct write lithography tool before we discovered that it was not a viable lithography tool for producing low-loss waveguides. However, upon etching, the micromasking effect that was appearing for the 0.42 μm -thick Ge was significantly amplified due to the thicker $\sim 2 \mu\text{m}$ -thick Ge, and the wafers were visibly damaged, as seen in Figure 3-12.

Another lot of the same wafers, regrown due to the initial surface defects, was received and had much fewer imperfections, as seen in Figure 3-14. Additional waveguides fabricated on these wafers should also transition to using e-beam lithography,

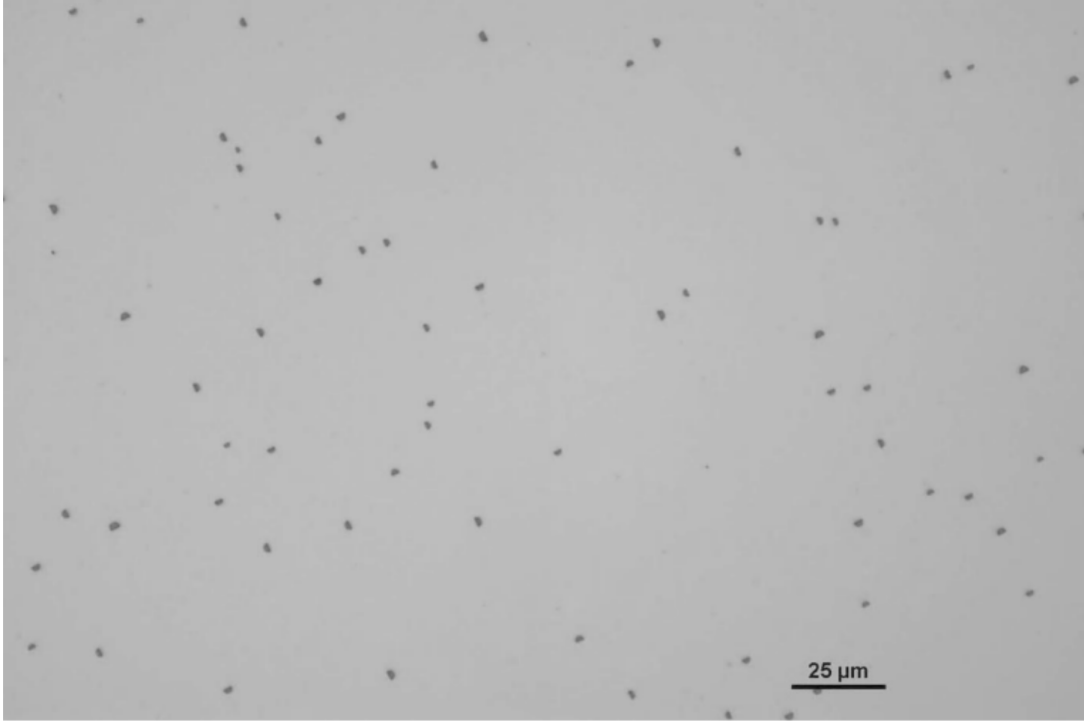


Figure 3-11: Wafer 1 with surface defects

as the issues with sidewall angle and roughness on the other set of waveguides will still be present with these wider, thicker waveguides.

3.3 3 μm Wavelength Loss Measurement Setup

The loss measurement process began with the initial direct write lithography fabricated wafer. The laser for this testing was the Firefly High Power A pulsed laser. We were unable to couple in due to the poorly cleaved facets, but improvements were made upon the existing setup to bring the optical path length into a single plane.

The current loss measurement setup for the 3 μm wavelength laser can be seen in Figure 3-15. The beam is directed by two mirrors to redirect the laser, a neutral density filter to reduce the laser power during alignment, an aspheric lens to focus the light into the waveguide, and a calcium fluoride (CaF_2) lens to collimate the beam. All lenses and the chip are on translation stages. There is a visible camera atop the



Figure 3-12: 4 separate etch steps were carried out to try to minimize any wafer over-heating. These images show the dose testing after the first and last etch, respectively, on 2 μm -thick Ge. The micromasking effect's damage is especially evident on the final result where almost the entire wafer has turned into "black-Si/Ge."

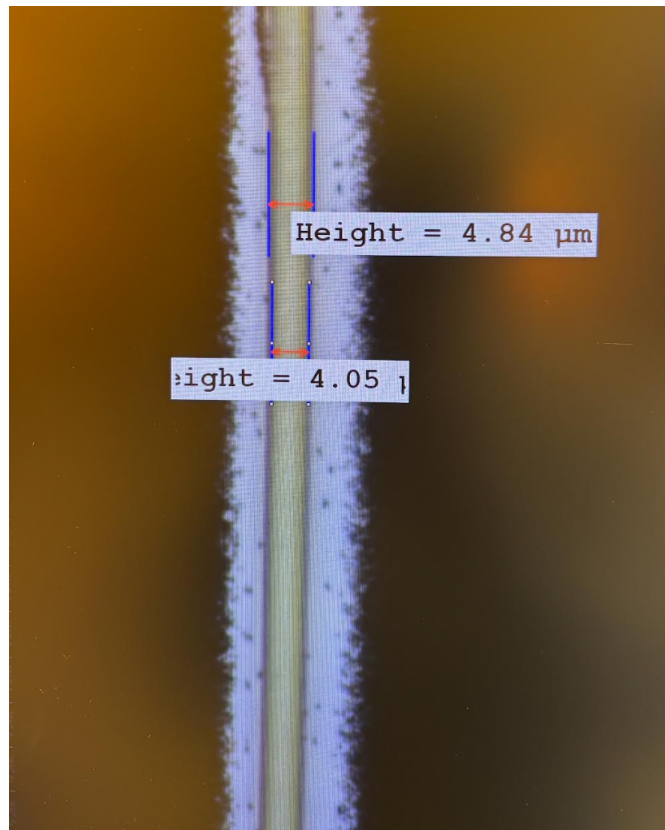


Figure 3-13: Waveguide next to Ge/Si pillars under microscope

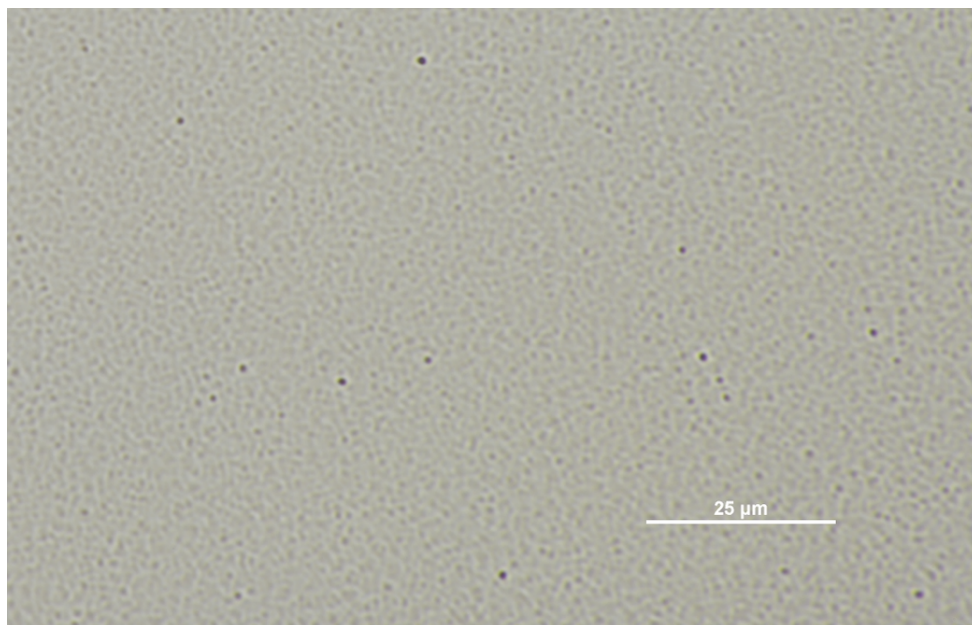


Figure 3-14: Significantly improved wafer lot for 2 μm thick Ge on Si substrate

photonic chip that is aligned with the mid-IR camera using the corner of the chip where there are recognizable light defraction patterns. This camera setup is used to orient the photonic chip with the laser for coupling in.

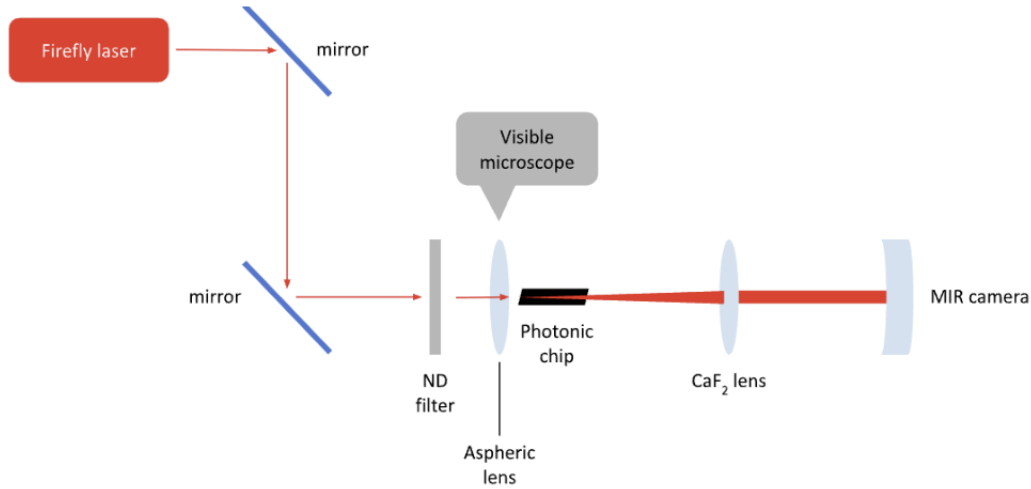


Figure 3-15: The loss measurement setup for the 3 μm wavelength laser

3.4 Summary and Next Steps

To conclude, we have modeled the largest strip waveguide dimensions that only support single TE mode propagation for two available lasers. For the 3 μm wavelength Firefly laser, a wafer from MIT Lincoln Labs had already been acquired and had around 0.5 μm thick Ge. Further imaging revealed that the Ge layer was actually 0.42 μm thick, which had been within the range supported in the modeling process since a slightly inaccurate thickness was expected. The largest width for a 0.42 μm thick Ge layer on Si substrate waveguide at 3 μm wavelength was 2.08 μm . Bend radii for this same wavelength were also modeled, and 250 μm was afforded to the loss measurement design for a bend radius that would not negatively impact getting reliable measurements or consuming excess wafer area. The waveguide design was created using Python scripts and varied the length while keep a consistent number of

bends in the waveguides in order to normalize loss measurements across the bends. A standard taper from a width of 15 μm down to 2.08 μm was included to facilitate the coupling in and out processes. For the longer 10.4-11.3 μm wavelength Daylight laser, we modeled the largest strip waveguides at different aspect ratios across the wavelength range. The modeling and an accompanying literature review revealed that an appropriate Ge layer thickness was around 2 μm , and wafers were requested from Lincoln Labs accordingly. This lot also included wafers with a SiO_2 ALD, an initial step towards exploring a new platform for better sensing capabilities.

We have also refined a process flow for direct write lithography. This was done using dose testing, changing the raster direction, and reducing the feature size. The result was a dose of 100 mJ/cm, a raster direction that followed along the straight sections of the waveguides, and a feature size of 600 nm, the smallest possible for the available direct write tool. An additional essential step was the HDMS prime for proper resist adhesion. We were able to consistently fabricate waveguides of approximately the correct width, but SEM imaging revealed that the resolution of direct write lithography was still too low, creating inconsistent waveguide widths and significant sidewall roughness. An effect that is likely micromasking was also observed with the existing chlorine etch recipe and flaws in the cleaving process were discovered when looking at the cleaved waveguide facets.

The continuation of this work is to fine-tune a process for fabricating waveguides using e-beam lithography for better resolution and with an HSQ mask to improve the resist selectivity for improved sidewall roughness. To avoid any micromasking effects in the etching, an HF dip immediately prior to the etch should be explored, and a concentration that removes native oxide and other impurities without removing the HSQ mask must be fine-tuned. Honing the cleaving process will also be crucial to getting repeatable loss measurements. This will consist of using a straight scribe and fully removing the resist (for an HSQ mask, this will consist of an HF dip) along with

other possible improvements. Future work may also consider other waveguide designs seen in literature, such as sub-wavelength grating (SWG) or rib waveguides. For practical applications in a foundry, a "universal" Ge layer thickness to be fabricated at different waveguide widths should be considered. This would support sensing applications on one chip across a wide range of wavelengths within the mid-IR. A universal thickness would also greatly simplify the fabrication process, as one mask and etch process could be used for different waveguides that support their respective wavelengths.

For example, at a Ge layer thickness of $1.7\text{ }\mu\text{m}$, single mode operation at $3\text{ }\mu\text{m}$ wavelength and $11.3\text{ }\mu\text{m}$ wavelength (the lower and upper bounds of our available lasers) are both supported. For the $3\text{ }\mu\text{m}$ wavelength, a strip waveguide width of $0.6\text{ }\mu\text{m}$ was found to support single TM mode polarization; for the $11.3\text{ }\mu\text{m}$ wavelength, a width of $6\text{ }\mu\text{m}$ was found for single TE mode polarization.

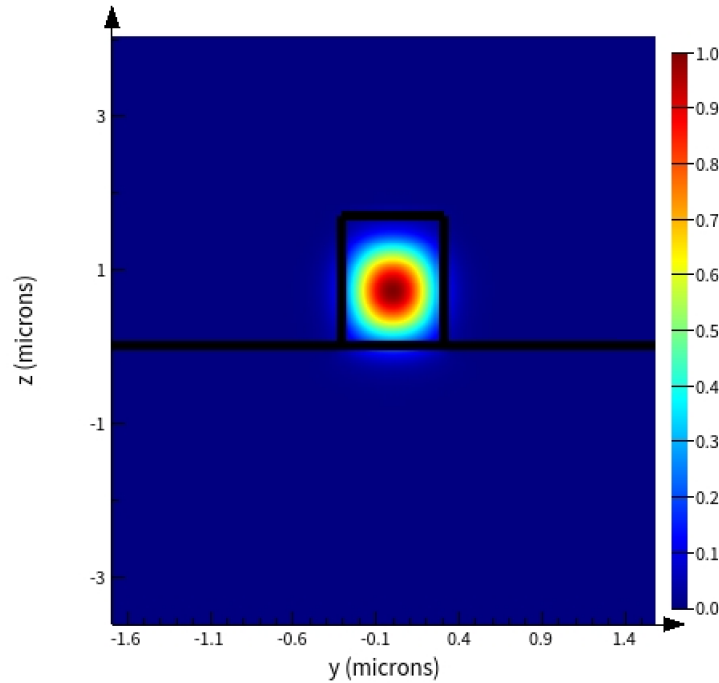


Figure 3-16: The single TM mode of a modeled strip waveguide with a width of $0.6\text{ }\mu\text{m}$ and a height of $1.7\text{ }\mu\text{m}$

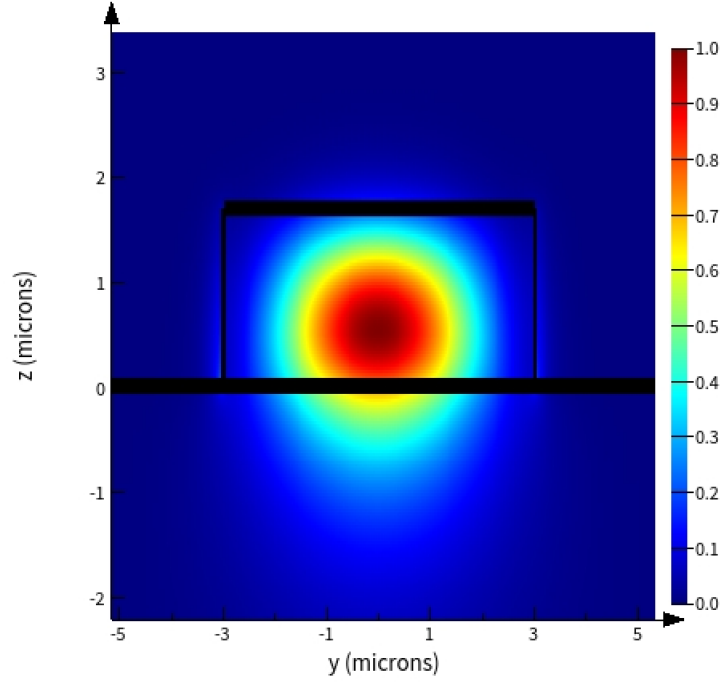


Figure 3-17: The single TE mode of a modeled strip waveguide with a width of $6\text{ }\mu\text{m}$ and the same height of $1.7\text{ }\mu\text{m}$ as Figure 3-15

Lastly, for the loss measurement setup, using overhead mid-IR alignment in place of the visible camera and/or chalcogenide fibers for more consistent coupling would streamline the testing process.

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